



“The Chemistry We Serve”



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兰亚化工
LANYACHEM

Partner for Innovative Silicone Solutions Worldwide

Global Presence | Fast Response | Local Service

COMPANY INTRODUCTION



Founded in 2007, LANYACHEM is a globally recognized enterprise specializing in the research, development, and production of fine chemical products, including organosilicon compounds, acetylacetone and its derivatives, as well as plastic additives. With certifications in ISO 9001 and ISO 14001, we are committed to maintaining the highest standards of quality, sustainability, and environmental responsibility.

Headquartered in Nanjing, China, our production facilities are strategically located in Hubei, ensuring efficient manufacturing and distribution. To better serve our global clients, we operate warehouses and product application support centers in Germany, the United States, South Korea, and India.

Guided by our motto, "THE CHEMISTRY, WE SERVE," LANYACHEM is dedicated to providing innovative solutions that drive progress in diverse industries. Leveraging state-of-the-art technology and a customer-centric approach, we are not just a supplier but a trusted partner in fostering sustainable growth and industrial advancement.

TEAM INTRODUCTION

SALES TEAM



LANYACHEM has a young and energetic sales team, widely service for the local enterprises. Sales team member has good professional background on fine chemicals and new industry, strictly trained on the technology, is a trust-based technical sales team. Sales team works closely with the customer, engaged in providing high value added products and service for the customer, creating value for the customer and making continuous development with the customer.

PRODUCTION TEAM

LANYACHEM does not only have advanced technology group and experienced production management, but also can guarantee the products are released after strict quality control before arriving at the customer. Now Lanya Chemical has skilled production staff and experienced on-site management, to ensure the products supplied to our customer are on the top grade.



R&D TEAM



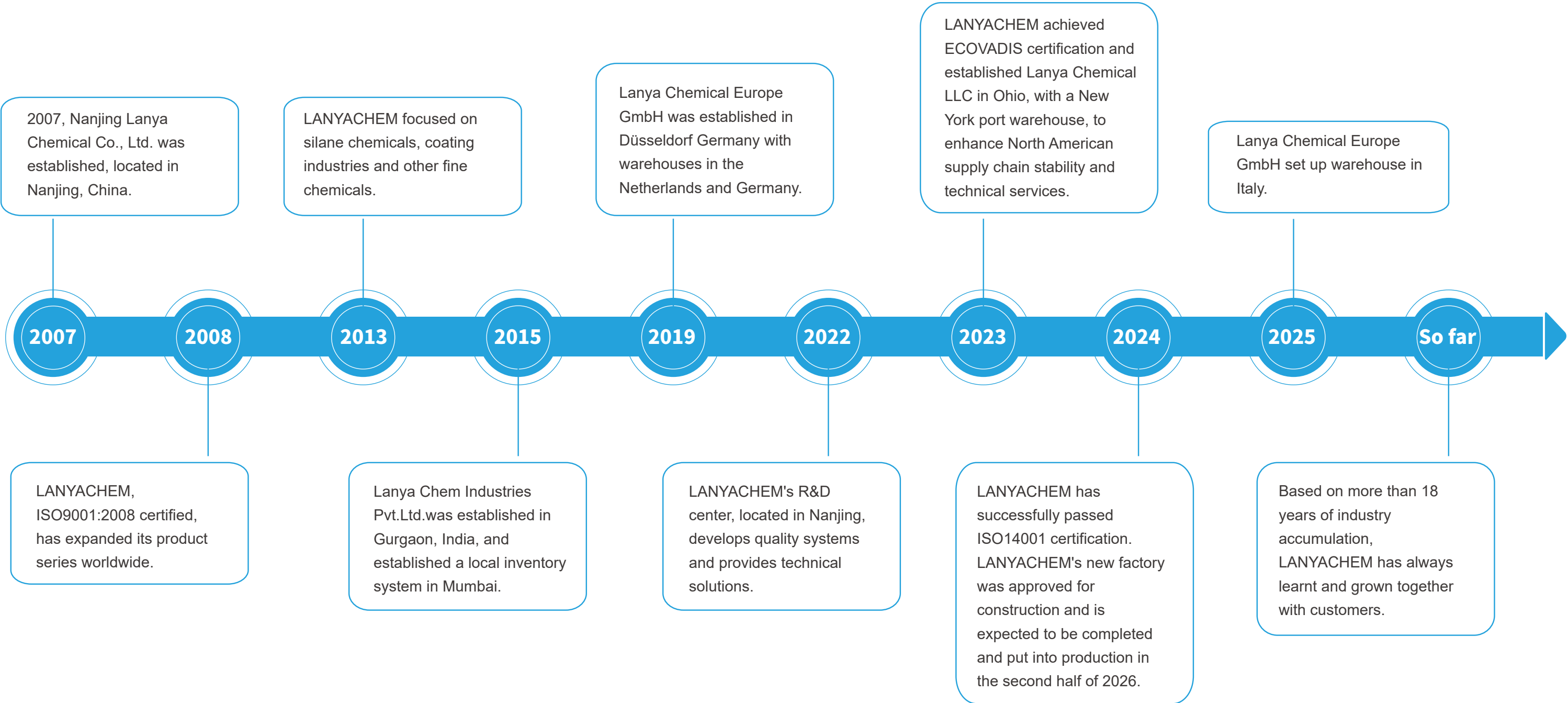
Based on the advantages of our R&D Centre, in order to guarantee the stability of the product quality and continues developing new products conforming to the plastic industry trend, LANYACHEM is always concentrating on the investment of the R&D. Our R&D team continuously follow up the production process and quality, continuously provide their suggestions on the improvement and engaged in developing new products to create more value for the customers.

APPLICATION TEAM

LANYACHEM's application team is always ready to communicate with the customers deeply, understand the currently exist issues or demand of the customer, so that they can share the perfect technology solutions. Cooperate with the sales team to investigate the customer's requirement, optimize the technology process of the usage, and expand the application field of the product, in order to create more added values for the customer.



COMPANY HISTORY



SALES MARKET MAP



- Warehouse: China (Nanjing), Netherlands (Rotterdam), India (Mumbai), Italy (Milan)
- ◆ Branch office: China (Nanjing & Guangzhou), Germany (Dusseldorf), South Korea (Inchon), India (New Delhi), USA (Columbus)
- ★ Distributor: Thailand, Indonesia, Bangladesh, the United States, Brazil, Mexico, Turkey, Australia, Algeria

Brand	Product Name	CAS No.	CAS Brand	Wacker	Dow-Corning	Shin-Etsu	Momentive	Evonik
Silane Monomer & Intermediates								
Silane Monomers								
YAC-MTCS	Methyltrichlorosilane	75-79-6	M1					
YAC-DMDCS	Dimethyldichlorosilane	75-78-5	M2					
YAC-TMCS	Trimethylchlorosilane	75-77-4	M3					
YAC-M1H	Methyldichlorosilane	75-54-7	MH					
YAC-M2H	Dimethylchlorosilane	1066-35-9						
YAC-PhTCS	Phenyltrichlorosilane	98-13-5						
YAC-DPhDCS	Diphenyldichlorosilane	80-10-4						
Silane Intermediates								
YAC-V150	Vinyltrichlorosilane	75-94-5						
YAC-C323	3-Chloropropyltriethoxysilane	5089-70-3			Z-6076	KBM-703	A-143	Dynasylan CPTMO
YAC-C313	3-Chloropropyltrimethoxysilane	2530-87-2			Z-6376	KBE-703		
YAC-C312	3-Chloropropylmethylmethoxysilane	18171-19-2						
YAC-C322	3-Chloropropylmethyldiethoxysilane	13501-76-3						
YAC-H113	Trimethoxysilane	2487-90-3						
YAC-H123	Triethoxysilane	998-30-1						
YAC-H112	Methyldimethoxysilane	16881-77-9						
YAC-H122	Methyldiethoxysilane	2031-62-1						
Silicone End-capping Agent								
YAC-HMM	1,1,3,3-Tetramethyldisiloxane	3277-26-7						
YAC-HMDSO	Hexamethyldisiloxane	107-46-0						
YAC-HMDS	Hexamethyldisilazane	999-97-3			Z-6079		A-166	Dynasylan HMDS
YAC-HPDS	Heptamethyldisilazane	920-68-3						Dynasylan HPDS
YAC-HMTS	1,1,1,3,5,5,5-Heptamethyltrisiloxane	1873-88-7						
YAC-ViTMDS	1,3-Divinyl-1,1,3,3-Tetramethyldisiloxane	2627-95-4						
YAC-ViDMES	Vinyldimethylethoxysilane	5356-83-2						

Brand	Product Name	CAS No.	CAS Brand	Wacker	Dow-Corning	Shin-Etsu	Momentive	Evonik
Cyclosiloxanes								
YAC-DMC	Dimethylcyclosiloxane	69430-24-6						
YAC-D3	Hexamethylcyclotrisiloxane	541-05-9						
YAC-D4	Octamethylcyclotetrasiloxane	556-67-2						
YAC-D5	Decamethylcyclopentasiloxane	541-02-6						
YAC-D4h	1,3,5,7-Tetramethylcyclotetrasiloxane	2370-88-9						
YAC-D4Ph	Octaphenylcyclotetrasiloxane	546-56-5						
YAC-V4	Tetramethyl Tetravinyl Cyclotetrasiloxane	2554-06-5						
YAC-D3N	Hexamethylcyclotrisilazane	1009-93-4						
YAC-D4N	Octamethylcyclotetrasilazane	1020-84-4						
Silane Coupling Agent								
α-Silane Series								
YAC-C123	Chloromethyltriethoxysilane	15267-95-5						
YAC-C112	Chloromethyl(methyl)dimethoxysilane	2212-11-5						
YAC-C111	Chloromethyl(dimethyl)methoxysilane	18143-33-4						
YAC-L1742	Methacryloxymethyltriethoxysilane	5577-72-0						
YAC-L323	Anilinomethyltriethoxysilane	3473-76-5						
YAC-L623	N-Cyclohexyl(aminomethyl)triethoxysilane	26495-91-0						
Sulfur Silane								
YAC-S619	Bis[3-(triethoxysilyl)propyl]tetrasulfide	40372-72-3			Z-6940	KBE-846	A-1289	Si-69
YAC-S619C	Mixturer of S619 (50%) and Carbon black (50%)	N.A.						
YAC-S629	Bis[3-(triethoxysilyl)propyl]disulfide	56706-10-6						
YAC-S629C	Mixturer of S629 (50%) and Carbon black (50%)	N.A.						
YAC-S313	3-Mercaptopropyltrimethoxysilane	4420-74-0	KH-590	GENIOSIL GF 70	Z-6062	KBM-803	A-189	Dynasylan MTMO/3201
YAC-S323	3-Mercaptopropyltriethoxysilane	14814-09-6	KH-580				A-1891	
YAC-S312	3-Mercaptopropylmethylmethoxysilane	31001-77-1						
YAC-S322	3-Mercaptopropylmethyldiethoxysilane	13818-38-7						

Brand	Product Name	CAS No.	CAS Brand	Wacker	Dow-Corning	Shin-Etsu	Momentive	Evonik
Amino Silane								
YAC-A110	3-Aminopropyltriethoxysilane	919-30-2	KH-550	GF 93	Z-6011	KBE-903	A-1100	Dynasylan AMEO
YAC-A111	3-Aminopropyltrimethoxysilane	13822-56-5	KH-540	GENIOSIL GF 96	Z-6610	KBM-903	A-1110	Dynasylan AMMO
YAC-A112	N-(2-Aminoethyl)-3-aminopropyltrimethoxysilane	1760-24-3	KH-792	GF 91	Z-6094	KBM-603	A-1120	Dynasylan DAMO
YAC-A122	N-(2-Aminoethyl)-3-aminopropyltriethoxysilane	5089-72-5						
YAC-A210	3-Aminopropylmethyldiethoxysilane	3179-76-8	KH-902		Z-6015	KBE-902	A-2100	Dynasylan 1505
YAC-A212	N-(2-Aminoethyl)-3-aminopropylmethyldimethoxysilane	3069-29-2	KH-602	GENIOSIL GF 95	Z-6436	KBM-602	A-2120	Dynasylan 1411
YAC-A161	3-Ureidopropyltrimethoxysilane	23843-64-3					A-11542	
YAC-A160	3-Ureidopropyltriethoxysilane (50% in Methanol)	23779-32-0			Z-6676	KBE-585	A-1160	Dynasylan 2201EQ
YAC-A113	3-[2-(2-Aminoethylamino)ethylamino]propyltrimethoxysilane	35141-30-1			AY43-009		A1130	Dynasylan TRIAMO
YAC-A312	3-Aminopropylmethyldimethoxysilane	3663-44-3						
YAC-DVA323	3-(1,3-Dimethylbutylidene)aminopropyltriethoxysilane	116229-43-7						
YAC-A115	N-(n-Butyl)-3-aminopropyltrimethoxysilane	31024-56-3						
YAC-A1151	N-(n-Butyl)-3-aminopropyltriethoxysilane	31024-56-3						
YAC-A616	Bis[3-(trimethoxysilyl)propyl]amine	82985-35-1						
YAC-A626	Bis[3-(triethoxysilyl)propyl]amine	13497-18-2						
YAC-AB313	3-Anilinopropyltrimethoxysilane	3068-76-6						
YAC-A4313	N,N-Diethylaminopropyltrimethoxysilane	41051-80-3						
YAC-A2315	N,N-Dimethylaminopropyltrimethoxysilane	2530-86-1						
YAC-A2322	N-(2-Aminoethyl)-3-aminopropylmethyldiethoxysilane	70240-34-5						
YAC-A2312	N-(2-aminoethyl)-3-aminoisobutylmethyldimethoxysilane	70240-34-5						
Isocyanato Silane								
YAC-IC313	3-Isocyanatopropyltrimethoxysilane	15396-00-6		GENIOSIL GF 40			A-Link 35	
YAC-IC323	3-Isocyanatopropyltriethoxysilane	24801-88-5		GENIOSIL GF 41		KBE-9007	A-Link 25	
Alkyl Silane								
YAC-TEOS28	Tetraethyl Orthosilicate 28	78-10-4		SILICATE TES 28	Z-6697	KBE-04	TEOS	Dynasylan A
YAC-TEOS32	Tetraethyl Orthosilicate 32	78-10-4						

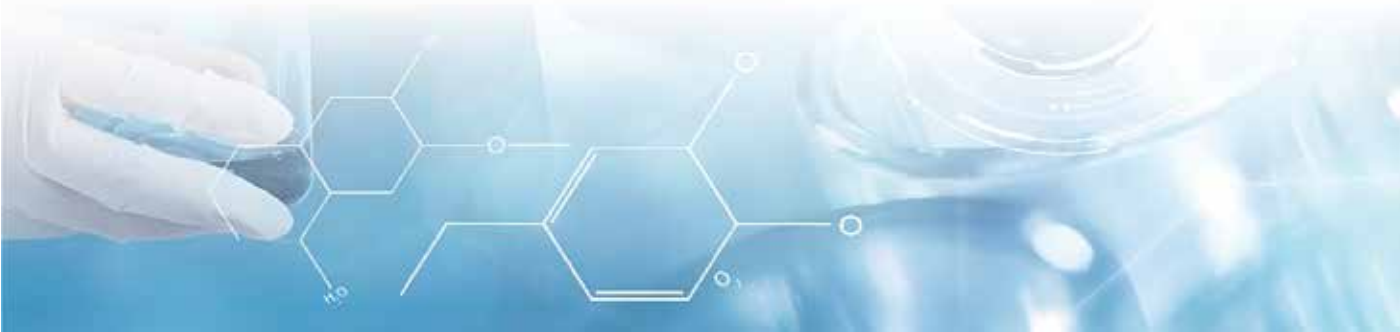
Brand	Product Name	CAS No.	CAS Brand	Wacker	Dow-Corning	Shin-Etsu	Momentive	Evonik
YAC-TEOS40	Tetraethyl Orthosilicate 40	78-10- 4		SILICATE TES 40			TEOS-40	Dynasylan 40
YAC-TMOS51	Tetramethyl Orthosilicate 51	681-84-5		GENIOSIL TM				Dynasylan M
YAC-TMOS56	Tetramethyl Orthosilicate 56	681-84-5						Dynasylan M
YAC-TPOS	Tetrapropyl Orthosilicate	682-01-9						Dynasylan P
YAC-N113	Methyltrimethoxysilane	1185-55-3			Z-6070	KBM-13	A-1630A	Dynasylan MTMS
YAC-N123	Methyltriethoxysilane	2031-67-6			Z-6370	KBE-13	A-162	Dynasylan MTES
YAC-N313	n-Propyltrimethoxysilane	1067-25-0						
YAC-N323	n-Propyltriethoxysilane	2550-02-9						
YAC-IN423	iso-Butyltriethoxysilane	17980-47-1						
YAC-N813	n-Octyltrimethoxysilane	3069-40-7			Z-6672			Dynasylan OCTMO
YAC-N823	n-Octyltriethoxysilane	2943-75-1			Z-6341		A-137 Y-9187	Dynasylan OCTEO
YAC-IN823	iso-Octyltriethoxysilane	35435-21-3						
YAC-IN813	iso-Octyltrimethoxysilane	34396-03-7						
YAC-N812	n-Octylmethyldimethoxysilane	85712-15-8						
YAC-N822	n-Octylmethyldiethoxysilane	2652-38-2						
YAC-N1213	n-Dodecyltrimethoxysilane	3069-21-4				KBM-3103C		
YAC-N1222	n-Dodecylmethyldiethoxysilane	60317-40-0						
YAC-N1223	n-Dodecyltriethoxysilane	18536-91-9						
YAC-N1613	n-Hexadecyltrimethoxysilane	16415-12-6						Dynasylan 9116
YAC-N1612	n-Hexadecylmethyldimethoxysilane	N.A.						
YAC-N1623	n-Hexadecyltriethoxysilane	16415-13-7						
YAC-N1813	n-Octadecyltrimethoxysilane	3069-42-9						
YAC-N1823	n-Octadecyltriethoxysilane	7399-00-0						
YAC-NT313	Methyltris(trimethylsiloxy)silane	17928-28-8						
YAC-NT412	Tetrakis(2-methoxyethoxy)silane	2157-45-1						
YAC-BTES	1,2-Bis(triethoxysilyl)ethane	16068-37-4						
YAC-BTMS	1,2-Bis(trimethoxysilyl)ethane	18406-41-2						

Brand	Product Name	CAS No.	CAS Brand	Wacker	Dow-Corning	Shin-Etsu	Momentive	Evonik
Phenyl Silane								
YAC-P13	Phenyltrimethoxysilane	2996-92-1			Z-6124	KBM-103	A-153	Dynasylan 9165
YAC-P23	Phenyltriethoxysilane	780-69-8			Z-9805	KBE-103		
YAC-P112	Methylphenyldimethoxysilane	3027-21-2						
YAC-P122	Methylphenyldiethoxysilane	775-56-4						
YAC-P212	Diphenyldimethoxysilane	6843-66-9			Z-6047	KBM-202SS		Dynasylan 6010
YAC-P130	Phenyltrimethylsilane	768-32-1						
YAC-P222	Diphenyldiethoxysilane	2553-19-7				KBE-202		
Acyl Silane								
YAC-O174	3-Methacryloxypropyltrimethoxysilane	2530-85-0	KH-570	GENIOSIL GF 31	Z-6030	KBM-503	A-174	DynasylanMEMO
YAC-O1741	3-Methacryloxypropylmethyldimethoxysilane	14513-34-9	KH-571		Z 6033	KBM-502		
YAC-O1742	3-Methacryloxypropyltriethoxysilane	21142-29-0			Z-6036	KBE-503	Y-11878 Y-9936	
YAC-O1743	3-Methacryloxypropylmethyldiethoxysilane	65100-04-1				KBE-502		
YAC-O174T3	3-(Methacryloyloxy)propyltris(trimethylsiloxy)silane	17096-07-0						
YAC-O1TAS	3-[Tris(acetoxy)silyl]propyl Methacrylate	51772-85-1						
Epoxy Silane								
YAC-O186	2-(3,4-Epoxy cyclohexyl)ethyltrimethoxysilane	3388-04-3			Z-6043	KBM-303	A-186	
YAC-O1861	2-(3,4-Epoxy cyclohexyl)ethyltriethoxysilane	10217-34-2					CoatOSil 1770	
YAC-O187	3-Glycidoxypropyltrimethoxysilane	2530-83-8	KH-560	GENIOSIL GF 80	Z-6040	KBM-403	A-187	Dynasylan GLYMO
YAC-O1871	3-Glycidoxypropyltriethoxysilane	2602-34-8		GENIOSIL GF 82	Z-6041	KBE-403	A-1871 Y-15589	Dynasylan GLYEO
YAC-O1872	3-Glycidoxypropylmethyldiethoxysilane	2897-60-1			Z- 6042	KBE-402	CoatOSil 2287	
YAC-O1873	3-Glycidoxypropylmethyldimethoxysilane	65799-47-5			Z-6044			
Vinyl Silane								
YAC-V151	Vinyltriethoxysilane	78-08-0		GENIOSIL GF 56	Z-6518	KBE-1003	A-151	Dynasylan VTEO
YAC-V171	Vinyltrimethoxysilane	2768-02-7		GENIOSIL XL 10	Z-6300	KBM-1003	A-171	Dynasylan VTMO
YAC-V172	Vinyltris(2-methoxyethoxy)silane	1067-53-4		GENIOSIL GF 58			A-172	Dynasylan VTMOEO

Brand	Product Name	CAS No.	CAS Brand	Wacker	Dow-Corning	Shin-Etsu	Momentive	Evonik
Silane Polymer								
Silane Oligomer								
YAC-AO5151	Amino Siloxane	N.A.						
YAC-AO5153	Amino Siloxane	N.A.						
YAC-AO5146	Diaminofunctional silane	N.A.						
YAC-AO5203	Amino Copolymer Oligomer Siloxane	N.A.						
YAC-OO200	Epoxy Oligomer Siloxane	N.A.						
YAC-OO7270	Acryloxy Multifunctional Oligomer Siloxane	N.A.						
YAC-SO9201	Mercapto Copolymer Siloxane	N.A.						
YAC-SO9000	Mercapto Copolymer Siloxane	N.A.						
YAC-VO1090	Vinyl Oligomer Siloxane	N.A.						
YAC-VO1098	Vinyl Oligomer Siloxane	N.A.						
YAC-VO1298	Vinyl Multifunctional Oligomer Siloxane	N.A.						
YAC-MP113	3-(Methoxypolyoxyethylene)trimethoxysilane	N.A.						
Silane Polymer								
YASIL-OH-	OH Polymer	N.A.						
YASIL-PDMS	Polydimethylsiloxane	N.A.						
YASIL-OH-PDMS-LINER	Hydroxy Terminated Linear Polysiloxane	N.A.						
YASIL-NC132	Methylpropyldichlorosilane hydrolysate	N.A.						
YASIL-6100	Functional siloxane	N.A.						
YASIL-TD51	Polydimethyl Silicone Dispersion Paste	N.A.						
Fluoro Silane								
YAC-F813	1H,1H,2H,2H-Perfluorooctyltrimethoxysilane	85857-16-5						
YAC-F823	1H,1H,2H,2H-Perfluorooctyltriethoxysilane	51851-37-7						Dynasylan F8261
YAC-F1013	1H,1H,2H,2H-Perfluorodecyltrimethoxysilane	83048-65-1				KBM-7803		
YAC-F1023	1H,1H,2H,2H-Perfluorodecyltriethoxysilane	101947-16-4						

Brand	Product Name	CAS No.	CAS Brand	Wacker	Dow-Corning	Shin-Etsu	Momentive	Evonik
Silane Crosslinking Agent								
Acetoxy Silane Series								
YAC-MTAS	Methyltriacetoxysilane	4253-34-3						
YAC-ETAS	Ethyltriacetoxysilane	17689-77-9						
YAC-VTAS	Vinyltriacetoxysilane	4130-08-9		GENIOSIL GF 62	Z-6075			
YAC-PTAS	Propyltriacetoxysilane	17865-07-5						
YAC-BDAC	Di-tert-butoxydiacetoxysilane	13170-23-5						Dynasylan BDAC
YAC-MPTAS8020	Methyl & Propyltriacetoxysilane 80/20	N.A.						
YAC-MPTAS4060	Methyl & Propyltriacetoxysilane 40/60	N.A.						
YAC-MPTAS3070	Methyl& Propyltriacetoxysilane 30/70	N.A.						
YAC-MPTAS2080	Methyl & Propyltriacetoxysilane 20/80	N.A.						
YAC-METAS8020	Methyl & Ethyltriacetoxysilane 80/20	N.A.						
YAC-METAS7030	Methyl & Ethyltriacetoxysilane 70/30	N.A.						
YAC-METAS5050	Methyl & Ethyltriacetoxysilane 50/50	N.A.						
YAC-METAS3070	Methyl & Ethyltriacetoxysilane 30/70	N.A.						
YAC-MTAS/BDAC	Methyltriacetoxysilane/Di-tert-butoxydiacetoxysilane	N.A.						
YAC-MBDM	Methyltriacetoxysilane/Di-tert-butoxydiacetoxysilane/DBTDL (72.5%MTAS+26.8%BDAC+0.7%DBTDL)	N.A.						
Ketoxime Silane Series								
YAC-MOS	Methyltris(methylethylketoxime)silane	22984-54-9			Z-9075			OX 10
YAC-VOS	Vinyltris(methylethylketoximino)silane	2224-33-1						OX 20
YAC-TOS	Tetra(methylethylketoxime)silane	34036-40-1						OX 30
YAC-POS	Phenyltris(methylethylketoxime)silane	34036-80-1						
YAC-MPKOS	Methyltris(methylpropylketoximo)silane	37859-55-5						
YAC-VPKOS	Vinyltris(methylpropylketoximo)silane	58190-62-8						
YAC-MIBKOS	Methyltris(methylisobutylketoximino)silane	37859-57-7						
YAC-VIBKOS	Vinyltris(methylisobutylketoxime)silane	56145-64-1						
YAC-DMDOS	Dimethyl di(methyl ethyl ketoxime)silane	37843-26-8						

Brand	Product Name	CAS No.	CAS Brand	Wacker	Dow-Corning	Shin-Etsu	Momentive	Evonik
YAC-MTOS9010	Methyl tris&Tetra(methyl ethyl ketoxime)silane 90/10	N.A.						
YAC-MTOS8020	Methyl tris & Tetra(methyl ethyl ketoxime)silane 80/20	N.A.						
YAC-VTOS6535	Vinyl tris & Tetra(methyl ethyl ketoxime)silane 65/35	N.A.						
YAC-VTOS8218	Vinyl Tris&Tetra(Methyl Ethyl Ketoxime)Silane 82/18	N.A.						
YAC-VMDAOS	Vinylmethoxy-di(acetonoximo)silane	797751-44-1						
YAC-VDMIBKOS	Methylvinyl di(methylisobutylketoximino)silane	156145-66-3						
YAC-TMIBKOS	Tetra(methylisobutylketoximino)silane	156145-62-9						
Catalyst								
Tin Catalyst								
DBTDL	Dibutyltin Dilaurate	77-58-7						
DBTDA	Dibutyltin Diacetate	1067-33-0						
DBTDO	Dibutyltin Oxide	818-08-6						
DBTDC	Dibutyltin Dichloride	683-18-1						
DBTBS	Dibutyltin Bis(2,4-Pentanedionate)	22673-19-4						
Platinum catalyst								
PCAT-	Platinum(0)-1,3-diethylene-1,1,3,3-tetramethyldisiloxane	68478-92-2						
Titanium Catalyst								
TAA	Titanium Acetylacetonate	14024-64-7						
TNBT	Tetrabutyl Orthotitanate	5593-70-4						
TSPP	Tetraisopropyl Orthotitanate	546-68-9						

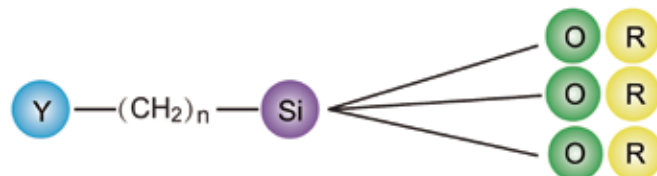


Organic Silane Coupling Agent Characteristics

Through chemical reaction or physical activity, organic silane coupling can be used to form a molecular bridge between the high molecular polymers and inorganic materials, to improve the physical characteristic, mechanic performance and electronic performance of the composite material.

Molecular Formula

Organic silane coupling agent is a organic silane molecular, which contains two type reactive group-inorganic reactive group and organic reactive group in one molecular.



Typical structure is: $Y(CH_2)_nSi(OR)_3$

In the structure, OR means a hydrolysis group - alkoxy, like methoxy, ethoxy, etc. Alkoxy can react with the added or residue water on the surface of the inorganic materials, hydrolyzed to hydroxyl, than the hydroxyl react with the hydroxyl on the surface of inorganic materials, condensation and one water molecular removed, then form a stable chemical bond.

In the structure, Y means an organic group, such as amino, methacryloxy, epoxy, vinyl, mercapto, etc. Different organic group is suitable for different organic polymers, it can take chemical reaction (thermosetting) or form physical wound, interpenetrating network system (thermoplastic) to form stable chemical bond.

Why Chosing Silane Coupling Agent?

When using glass fiber or mineral fillings to strengthen the organic polymers, the interface of polymers and inorganic material involves many complicated physical and chemical interactions, such interaction has closed relationship with the binding force, physical strength and insulation properties of the material. The main destructive factor on the effluence of the binding force is the water moves to the hydrophilic surface of the inorganic reinforced material. Water eroded the interface, damage the biding with the strengthen material. But the coupling agent treated material can form a waterproof bind between the inorganic and organic material. Silane coupling agent has special chemical and physical performance, it can not only enhance the binding strength, but also prevent the dissociation of the bind during the aging and usage. Coupling agent endow a stable binding ability between two different, repelled surfaces.

In the composite materials, choosing appropriate silane can greatly improve its bending strength (tensile strength and modulus), in the meantime, enhance the material's resistance to the moisture and other malignant environmental conditions.

The other advantage of organic silane coupling agent also includes:

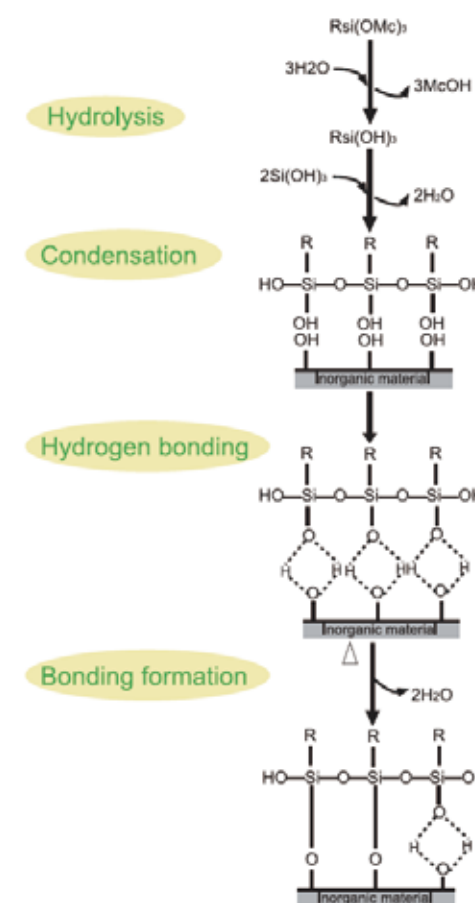
- ◆ Better wetting properties of resin
- ◆ Wet resistance and water removal agent
- ◆ Waterproof to constructions
- ◆ Better dispersion of mineral fillings
- ◆ More clear and transparent of the modified plastic

Reaction Mechanism of Silane to Inorganics

Silane coupling agent containing three inorganic reaction group (usually methoxy, ethoxy or acetyl) can better combine with the metal hydroxyl in most inorganic materials, especially in the structure containing Si, Al or heavy metals. By reacting with the added or residue water on the inorganic material surface, silyloxy can hydrolyzed to silanol, then these silanol can react with the metal hydroxyl on the inorganic surface, and formalkoxystructure and remove the water.

Silane can also bonded to each other, form a multi-molecular structure of organic silane agent. Usually, more than one layer or one layer silane will be used on the sureface. This will make a dense silyloxy layer on the inorganic material surface, and make the inorganic surface farther from the interface.

Mechnaisma of action onsilane coupling agents

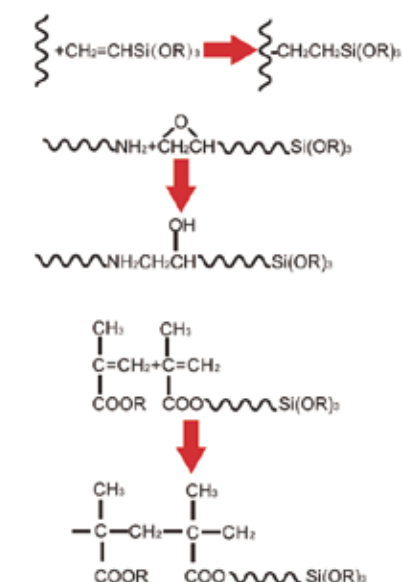


Reaction Mechanism of Silane to Polymers

The combination between the silane and organic polymer is very complicated. The reactivity of thermosetting polymers should match the reactivity of silanes. For example, epoxysilane or amino silane should be combined with epoxy resin; amino silane should be combined with phenolic resin; methacryloxy silane should be crosslinked with unsaturated polyester resin, like styrene. For the thermosetting polymers, which reactivity makes more important, the match of chemical structure will enhance the physical performance of the modified composites. Commonly, the reaction of silane and resin will have following three types:

- ◆ Graft
- ◆ Chemical Bond
- ◆ Copolymer

For thermoplastic polymers, the combination by silane coupling agent can be explained as a formulation of interdiffusion and interpenetrating network system in the interface region. So, for thermoplastic polymers, the compatibility of silane and resin is very important. The match on the solubility parameter of two materials will help to improve the possibility on getting the best performance of the composite materials.



Adhesives & Sealants

Silane can improve the adhesion of most silicone sealant and adhesives on the metal, glass, stone and other inorganic bases.

Benefit:

·Initial adhesion	·Chemical resistance	·Corrosion resistance
·Lasting adhesion	·Tear strength	·Packaging stability
·Moisture resistance	·Elongation at break	·Curing uniformity
·Temperature resistance	·Stress-cracking resistance	·Filler dispersion
·Adhesive on the filler and resin	·Strength of the substrate	



Plastics & Rubber

Silane can be used to bond the inorganic filler and pigment onto the polymers, optimize the production of PP, crosslink of PE and the production of thermoplastic vulcanizates (TPVS). Besides, it can also be used to produce Silica reinforced rubber, which can be used in manufacturing “green” tires.

Benefit:

·Improve toughness	·Wear resistance	·Corrosion resistance
·Easy to process	·Stress-cracking resistance	·Packaging stability
·Temperature resistance	·Low temperature performance	·Strength of the substrate
·Adhesive of filler and resin	·Electrical performance	·Filler dispersion



Composites

Silane can bring significant difference on glass fiber reinforced polymers. Glass fiber is very hydrophilic, can bring the water onto the interface. Withoutsilane treated on the surface, the adhesion between the glass fiber and theresin will become weaker and weaker, until failed finally.

Benefit:

·Improve toughness	·Protection and treatment of fiber bundles
·Easy to process	·Anti-cracking and glitches
·Wetting of glass fiber	·Low temperature performance
·Adhesion of glass fiber and resin	·Electrical performance
·Strength of composite material	·Temperature variability



Energy, Electrics & Electronics

Silane can improve the reliability of semiconductor encapsulation materials, flat-panel displays and other products. Silicone Fluids can be used as transformer oil, cable oil, damper oil, condenser oil; preventing adherence of toner to photocopier rolls, etc. Silicone rubber materials play a key role in electronics and power distribution.

Benefit:

·Improve battery conversion efficiency	·Good fire resistance
·Optimize semiconductor performance	·Increase product durability
·Dehumidification	·Improve safety performance
·Extend material service life	·High mechanical properties
·High temperature resistance	·High insulation performance



Coatings, Paints & Inks

Silane can form a covalent bond between the inorganic and organic compoundThe stability of Siloxane group (Si-O-Si) make this technology become a key partin the high performance painting and coating.

Benefit:

·Improve toughness	·Thermal stability and durability
·Dispersion of pigment and filler	·Mobility
·Water and chemical resistance	·UV resistance
·Improve adhesion	·Defoaming



Building, Construction & Infrastructure

Using silane to prepare the permeability repellent, it can be used in porous substrates, hole blocking sealant and assembled agent, and extend the shelf life of masonry, stone and concrete.

Benefit:

·Water resistance	·Strong breathability
·Easy construction	·Maintain the surface appearance
·Persistent	·Weathering resistance
·Strong permeability	





How to Choose Silane Coupling Agent?

The hydrolysis rate of the silane coupling agent depends on the hydrolyzable functional group Si(OR)₃, and reaction activity with organic polymer depends on the organic functional group Y. Thus, it is essential to choose a suitable silane coupling agent for different kinds of substrate or the processing target.

Match the organic functional groups of the silicon atom and the polymer resin which will be combined can guide what kind of silane coupling agent can be used in a specific application. The organic group on the base on silane can be a reactive organic group (such as an organic functional group) or non-reactive organic group. These groups may be hydrophobic or hydrophilic with characteristics of different thermal stability.

Form 1 Characteristics of Different Organic Substituent on Silanes.

Organic Silane	
Y	The characteristics of the Y
Methyl	Hydrophobic, lipophilic
Phenyl	Hydrophobic, lipophilic, thermal stability
Long-chain Alkyl	The strong hydrophobic, lipophilic
Amine	Hydrophobic, organic reactions
Epoxy	Hydrophobic, organic reactions
Methacryloxy	Hydrophobic, organic reactions

Because of the different organic structure, the solubility parameter of the group is different to some extent, this will affect the interpenetration between polymer system and the silicone system for surface treatment. The select of silane should include the matching chemical reaction, dissolution characteristics and structural characteristics, even the thermal stability of the organic silane with the same parameters in the polymer structure.

Form 2 Solubility Parameter of Some Silanes

Product name	Functional groups	(RO) ₃ Si -Y	-Y
YAC-A110	Amino	8.56	9.86
YAC-A112	Amino	9.00	10.24
YAC-O174	Acyloxy	8.66	9.48
YAC-O186	Epoxy	8.58	9.38
YAC-O187	Epoxy	8.49	9.29
YAC-O1871	Epoxy	8.51	9.29
YAC-V171	Vinyl	7.49	7.00
YAC-V151	Vinyl	7.76	7.00
YAC-S313	Mercapto	8.49	9.57

Different type functional group of the silane coupling agent is suitable for different type polymers. According to the correlation of chemical structure characteristics between the silane coupling agent and polymer, form 3 can help you make a choice.

Form 3 Guidelines on Selecting Silanes.

	Fuctional groups								
		Amino	Epoxy	Sulfenyl	Mercapto	Acyloxy	Vinyl	Carbamido	Isocyanate group
Thermosetting resin	Polymer								
	Furan resin	⊙	○						○
	Unsaturated resin		○			⊙	○		
	Diallyl phthalate		○			○	○		
	Polyimide	○	○					○	○
	Polyurethane	○	○		○			○	⊙
	Epoxy resin	⊙	⊙						○
	Phenolic resin	⊙	○					○	○
Thermoplastic resin	ABS	○	○		○	⊙			○
	PET/ PBT	○	○						⊙
	Polyurethane	○	○		○			○	⊙
	Nylon	⊙	○					⊙	○
	Polycarbonate	○	○			○			○
	PVC	⊙	○		○				
	Acrylic acid	⊙	○		○	○			
	Polystyrene	⊙	○		○	⊙			
	PP	○	○		○				
	PE	○	○		○				
Emulsion	Acrylic Emulsion		□			⊙	⊙		
	Latex					⊙	⊙		
	Water-based polyurethane	□	□						
Elastomer / Rubber	Polyurethane rubber	○	○	○	⊙			○	
	Thiokol rubber	○	○	○	⊙				
	Butyl rubber	○	○						
	Neoprene	○		○	⊙				
	Chloropropionyl(ether)rubber	○		○					
	NBR	○	○		⊙				
	SBR			○	⊙				
	EPDM rubber	○		○	○	⊙	○		
	Polyisoprene rubber				○				
	Polybutadiene rubber				○				

⊙ Preferred ○ Applicable □ Only applies to some special silane

Treatment on Inorganics

Using silane coupling agent to treat inorganic materials, it's also called surface treatment. After the preparation of the solution, it can be used by dipping, spray, brushing or some other method. Usually, dipping is adopted for bulk material, granular material and glass fiber, spray is adopted for granular materials; if monolithic coating on the substrate required, brushing method is adopted.

Usually silane is used as diluted water solution. Silane coupling agent contains the hydrolyzable alkoxy, which can react with the inorganic materials, when these groups hydrolyzed, silanol is formed and can be reacted with hydroxyl group on the surface of inorganic material to form bond. Hydrolyzable alkoxy includes methoxy and ethoxy group. Methoxy and ethoxy group has different hydrolysis property, suitable for different application.

Preparation of Silane Solution

◆ Silane coupling agent concentration

Silane coupling agent is usually used under diluted solution, concentration is around 0.1~2.0%, the solvent is usually water, alcohol or water-alcohol mixture.

◆ PH adjustment

Except amino silane, if the silane coupling agent is not easily water soluble, can use 0.1~2.0% acetic acid water solution to adjust the pH value to weak acid. Amino silane, please refer to next page.

◆ Silanes, which are unsuitable to prepare the water solution

Long-chain alkyl and phenyl silane are not soluble in water, can be diluted with organic solvent or add 0.1~0.2% non-ionic surfactant, then add water and prepare it to water emulsion. For chloro and acetoxysilane, there's serious condensation during the hydrolysis, and not suitable to prepare the water solution or water-alcohol solution.

◆ Dropwise

Add the silane coupling agent when stirring. Shake it quickly, be cautious not to drop too quickly, to avoid spill. If add all in one lot, it's easily to have problem on weak dispersion or occur colloidal particles. After dropping silanes, should continuously stir for 30-60min, until the solution becomes transparent and shows totally hydrolyzed.

◆ Filter the water solution

If you find any un-soluble suspended solids in the solution, it's better to remove it. If it's in the is recommended.

The Effectiveness of Silane Coupling Agent to Various Inorganic Materials

■ The effectiveness of silane coupling agent to various inorganic materials

Effectiveness	Inorganic materials
Very Effective	Glass, Silica, Aluminium oxide
Effective	Mica, Clay, Al, Aluminium hydroxide
Slightly Effective	Asbestos, TiO ₂ , ZnO, Fe ₂ O ₃
Almost no Effect	Graphite, Carbon black, Calcium carbide

The alkoxy (like methoxy or ethoxy), alkyl in the silane coupling agent can react with water to form silanol. These silanol groups are not stable, and can be easily condensed to siloxane bond after the time going, finally caused the gelation. Usually, silanol group is not stable in water solution, but such stability can be improved in weak acid solution. But in the water solution of amino silanes, because of interactions, the silane becomes very stable. Solution on improving the stability of non-amino silanes includes adjustment of solution PH value (PH=4~5), make it combined with alcohol, and store it in room temperature or low temp

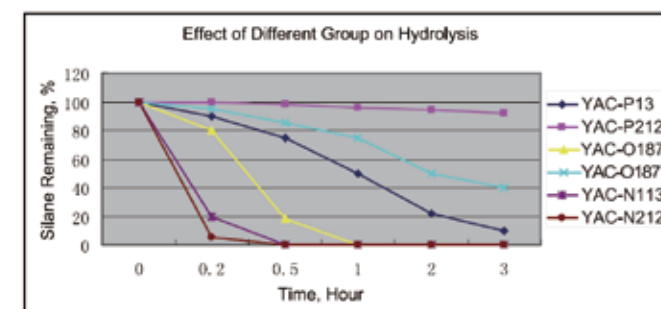
Form 4 Water solubility and stability of common silanes under optimal pH value:

Brand	Solubility(pH of watersolution)	Shelf life
YAC-A110	◎ (10.0)	30 Days
YAC-A112	◎ (10.0)	30 Days
YAC-O174	O (4.2)	1 Day
YAC-O186	O (4.0)	30 Days
YAC-O187	◎ (5.3)	30 Days
YAC-O1871	O (4.0)	10 Days
YAC-V171	O (3.9)	10 Days
YAC-V151	O (3.9)	10 Days
YAC-S313	O (4.0)	1 Day

◎ 1% silane water solution can be prepared without adjust the water PH value

O 0.1% silane water solution can be prepared with adjust the water PH value

* Upon different applied condition and purpose, the shelf life might be different



Treatment on Organics

This method has lower efficient then pre-treatment of inorganic materials in a certain extent, but it's lowered down the operation and raw material charges, still widely applied to industrial production.

Finishing the Mixture Method

In this method, silane coupling agent is added into the inorganic materials before mixing the organic and inorganic materials. Because amino silane has effect on solidification, add amino silane into the epoxy resin or phenolic resin can increase the material's viscosity or even solidification, so, please use it as soon as possible.

Masterbatch Method

This method is using silane coupling agent and small amount of organic material to produce masterbatch, then add the masterbatch to other materials.

React with Organic Silane (Copolymer)

This method has lower efficient then pre-treatment of inorganic materials in a certain extent, but it's lowered down the operation and raw material charges, still widely applied to industrial production.

